

L9a: Introduction to Derivatives and European Option Pricing

CHEME 5660 Cornell University

Copyright Jeffrey Varner 2026. All Rights Reserved

Disclaimer and Risks

Training only. This content is for training and informational purposes only. The teaching team does not make, give, or endorse any offer or solicitation to buy or sell securities or derivative products, or provide investment or trading advice or strategy.

Risk. Trading involves risk and is generally not recommended for those with limited resources, experience, or a low risk tolerance. Past performance does not guarantee future success.

Responsibility. You are responsible for your investment decisions based on your financial circumstances, objectives, risk tolerance, and liquidity needs.

Objectives for Today

A Kalshi contract costs less than a dollar today and pays one dollar only if an event occurs. What is a payment that may never arrive worth today?

Today's concepts

- Define and classify derivatives: futures, forwards, swaps, and options.
- Analyze option payoffs and profits: long and short calls and puts at expiration, breakevens, and loss bounds.
- Price European options contracts: the premium makes the expected net present value zero under the risk-neutral measure, the Black–Scholes–Merton formulas compute it, and put–call parity links calls and puts.

Lectures and Examples

Lecture:
CHEME-5660-L9a-Lecture-IntroductionToDerivatives-BSM-Fall-2026.ipynb

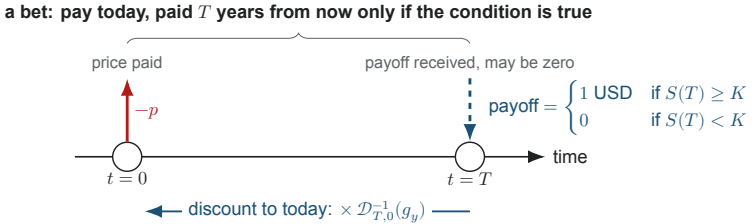
Example 1:
CHEME-5660-L9a-Example-SingleContractPayoffProfit-Fall-2026.ipynb

Example 2:
CHEME-5660-L9a-Example-BSM-Premium-Fall-2026.ipynb

The first example computes payoff and profit for single long and short call and put contracts. The second computes European premiums with Black–Scholes–Merton and Monte Carlo simulation, then varies the strike.

Concept Review: A Bet on the Future

Suppose we pay p today for a contract that pays 1 USD in T years if $S(T) \geq K$, and nothing otherwise.



Discounting the payoff at the benchmark growth rate g_y gives a random NPV:

$$\text{NPV}_0 = -p + \mathcal{D}_{T,0}^{-1}(g_y)\, V(S(T)) = \begin{cases} -p + e^{-g_y T} & \text{if } S(T) \geq K \text{ (true)} \\ -p & \text{if } S(T) < K \text{ (false)} \end{cases}$$

Expected NPV of the Bet

Under our beliefs about $S(T)$, the expected NPV is:

$$\mathbb{E}[\text{NPV}_0] = (-p + e^{-g_y T}) \underbrace{\mathbb{P}(S(T) \geq K)}_{\text{true}} + (-p) \underbrace{\mathbb{P}(S(T) < K)}_{\text{false}} \qquad \text{weight by probability}$$

$$= -p \underbrace{[\mathbb{P}(S(T) \geq K) + \mathbb{P}(S(T) < K)]}_{=1} + e^{-g_y T} \mathbb{P}(S(T) \geq K) \qquad \text{collect the } p \text{ terms}$$

$$= -p + e^{-g_y T} \underbrace{\mathbb{P}(S(T) \geq K)}_{\text{true}} \qquad \text{probabilities sum to one}$$

The bet has a non-negative expected NPV when $p \leq e^{-g_y T} \mathbb{P}(S(T) \geq K)$.

The bet is a derivative contract: its payoff depends on the future price of another asset. Every buyer pays one market price, yet buyers with different beliefs value the same contract differently.

How is the market price set?

Company Profile: Cboe Global Markets

Cboe opened on April 26, 1973, as the Chicago Board Options Exchange, the world's first exchange for listed options. Black and Scholes published their pricing formula the same year.

- Listed options: standard strikes, expirations, and contract sizes, cleared by the Options Clearing Corporation (OCC).
- SPX options: European-style and cash-settled. Standard stock and ETF options are American-style.
- VIX: the market's expected 30-day volatility of the S&P 500, read from SPX option premiums.
- Event contracts: since June 2026, binary options on the Mini-S&P 500 that pay 100 USD or nothing.

Today's question: where does a listed premium come from?

What Is a Derivative?

A derivative is a contract between a buyer and a seller to transact an asset at a set price on a future date. In L1b's language, it is an abstract asset whose cash flows depend on the underlying asset.

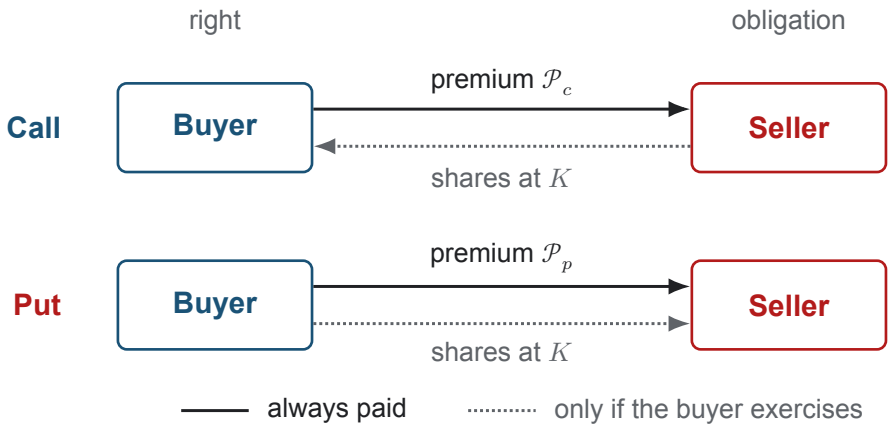
- Futures and forwards: agreements to buy or sell later at a set price. Futures are exchange-traded, and forwards trade over the counter.
- Swaps: one party exchanges an asset's value or cash flows for another's, over the counter.
- Options: the buyer holds the right, but not the obligation, to buy (call) or sell (put) at a set price. Options are exchange-traded.
- Event contracts: pay a fixed amount, usually one dollar, if an event occurs and nothing otherwise. Kalshi lists them.

How Big Is the US Options Market?

- 2025: a record of about 15.2 billion listed contracts, up about 24% from the 2024 record of 12.2 billion (OCC).
- Busiest day: about 110 million contracts on October 10, 2025 (Cboe).
- 2026: through September, about 70.6 million contracts per day, against 60.6 million for all of 2025, and about 13.2 billion contracts in total (OCC).
- Notional: option flow topped 3 trillion USD per day in January 2025, about 5 times the 2024 daily notional of U.S. cash equities (Cboe).

Sources: the lecture notebook's market table, with dates and links.

Call and Put Options Contracts



The Business Case: Who Risks What

Per share, at expiration:

Position	Maximum profit	Maximum loss
Long call	unbounded	the premium $\mathcal{P}_c$
Short call	the premium $\mathcal{P}_c$	unbounded
Long put	$K - \mathcal{P}_p$, if the share price falls to zero	the premium $\mathcal{P}_p$
Short put	the premium $\mathcal{P}_p$	$K - \mathcal{P}_p$, if the share price falls to zero

Only the uncovered short call can lose without bound. A short put's loss is large but bounded, because the share price cannot fall below zero. Covered calls and cash-secured puts hold the shares or cash for 100 shares per contract.

Call and Put Payoffs

A call gives the owner the right, but not the obligation, to buy 100 shares at the strike price K by the expiration date. A put gives the right to sell them. With share price $S(T)$ at expiration, the payoffs per share are:

$$\boxed{\begin{aligned} V_c(K, S(T)) &= \max(S(T) - K,\ 0) \\ V_p(K, S(T)) &= \max(K - S(T),\ 0) \end{aligned}}$$

The call is exercised only when $S(T) > K$, and the put only when $S(T) < K$. Both payoffs are nonnegative because the buyer may decline. The seller charges a premium, $\mathcal{P}_c$ for a call and $\mathcal{P}_p$ for a put.

Profit and Breakeven

The buyer's profit per share is the payoff minus the premium, and the seller's is the premium minus the payoff:

$$P_c^{\text{buyer}} = V_c(K, S(T)) - \mathcal{P}_c \qquad P_c^{\text{seller}} = \mathcal{P}_c - V_c(K, S(T))$$
$$P_p^{\text{buyer}} = V_p(K, S(T)) - \mathcal{P}_p \qquad P_p^{\text{seller}} = \mathcal{P}_p - V_p(K, S(T))$$

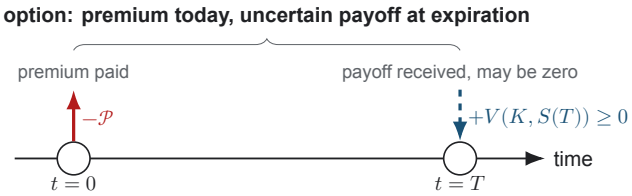
Setting the buyer's profit to zero gives the breakeven share price at expiration:

$$S_{\text{BE}} = K + \mathcal{P}_c \text{ (call)}, \qquad S_{\text{BE}} = K - \mathcal{P}_p \text{ (put)}$$

Above the call breakeven, and below the put breakeven, the buyer profits and the seller loses.

Example 1: CHEME-5660-L9a-Example-SingleContractPayoffProfit-Fall-2026.ipynb

Options as Abstract Assets



The buyer pays the premium $\mathcal{P}$ today and receives the payoff $V(K, S(T)) \geq 0$ at expiration, so the net present value at $t = 0$ is:

$$\text{NPV}_0 = -\mathcal{P} + \mathcal{D}_{T,0}^{-1}(g_y) \cdot V(K, S(T))$$

The profit $V - \mathcal{P}$ adds dollars from different dates. In today's dollars, the call buyer breaks even at $K + \mathcal{P}_c \cdot \mathcal{D}_{T,0}(g_y)$ rather than $K + \mathcal{P}_c$.

The European Premium

When trading shares and the risk-free asset can replicate an option's payoff, no arbitrage fixes its premium as the portfolio's cost. Under the corresponding risk-neutral measure $\mathbb{Q}$, this gives zero expected NPV:

$$\mathbb{E}_{\mathbb{Q}}(\mathrm{NPV}_0) = 0 \implies \boxed{\mathcal{P}^{\mathsf{EU}} = \mathbb{E}_{\mathbb{Q}}(\mathcal{D}_{T,0}^{-1}(g_y) \cdot V(K, S(T)))}$$

With frictionless replication, any other premium creates a risk-free profit:

- Too low: buy the option and sell a portfolio with the same payoff, built from shares and the risk-free asset.
- Too high: sell the option and buy the portfolio.

What $\mathbb{Q}$ Means

$\mathbb{Q}$ is a pricing rule, while the real-world measure $\mathbb{P}$ still describes how often outcomes occur:

- The share: under $\mathbb{Q}$, the expected share price grows at the risk-free rate g_y (no dividends), so $\mathbb{E}_{\mathbb{Q}}(\mathcal{D}_{T,0}^{-1}(g_y) \cdot S(T)) = S(0)$ and the share also has zero expected NPV.
- The option: one that expires worthless has scaled NPV $\rho_T = -1$. For a call, that happens with real-world probability $\mathbb{P}(S(T) \leq K)$ (L5a).
- The event contract: a one-dollar contract costs $p = \mathcal{D}_{T,0}^{-1}(g_y) \cdot \mathbb{Q}(\text{event})$, a discounted $\mathbb{Q}$-probability that can differ from $\mathbb{P}(\text{event})$.

A stock-price event, such as the bet in the Concept Review, takes its $\mathbb{Q}$ from the replicating model, but a general Kalshi event has no replicating portfolio, so choosing its pricing probability requires additional assumptions.

Why Is the Bet So Cheap?

Take a one-year bet that a share finishes at or above today's price, with 30% volatility, a 6% real-world mean growth rate, and a 4% risk-free rate:

- Real world: the bet pays off 58% of the time.
- A client with 50,000 USD and relative risk aversion 3 would pay 0.56 USD for it.
- The market charges only 0.47 USD.

Where did the other 8 cents go? The answer depends on when the bet pays, not on how risky the bet is on its own.

Optional: contingent claims notebook, last section.

Exercise Styles

The exercise style changes the premium:

- European: exercise only on the expiration date T.
- American: exercise at any time up to T.
- Bermudan: exercise only on specified dates up to T.

The American holder exercises at a time $\tau \leq T$ chosen using only the information available at that time (a stopping time):

$$\mathcal{P}^{\mathsf{AM}} = \sup_{\tau \leq T} \mathbb{E}_{\mathbb{Q}}\Big(\mathcal{D}_{\tau,0}^{-1}(g_y) \cdot V(K, S(\tau))\Big) \geq \mathcal{P}^{\mathsf{EU}}$$

Black–Scholes–Merton Premiums

Assume a non-dividend-paying share follows GBM with constant volatility σ and risk-free rate g_y, and frictionless continuous trading in the share and risk-free asset. The European call and put premiums are:

$$\boxed{\begin{aligned}\mathcal{P}_c &= N(d_+)\cdot S(0) - N(d_-)\cdot K\cdot \mathcal{D}_{T,0}^{-1}(g_y)\\ \mathcal{P}_p &= N(-d_-)\cdot K\cdot \mathcal{D}_{T,0}^{-1}(g_y) - N(-d_+)\cdot S(0)\end{aligned}}$$

where N is the standard normal CDF, T is in years, and:

$$d_+ = \frac{1}{\sigma\sqrt{T}}\left[\ln\frac{S(0)}{K} + \left(g_y + \frac{\sigma^2}{2}\right)T\right], \qquad d_- = d_+ - \sigma\sqrt{T}$$

Merton and Scholes received the 1997 Nobel Prize in Economic Sciences. Fischer Black had passed away before the prize was awarded.

Put–Call Parity

For a call and a put with the same strike and expiration, the payoffs differ by $S(T) - K$ at every share price:

$$V_c(K, S(T)) - V_p(K, S(T)) = S(T) - K$$

Subtracting the put premium condition from the call premium condition, and using $\mathbb{E}_{\mathbb{Q}}(\mathcal{D}_{T,0}^{-1}(g_y) \cdot S(T)) = S(0)$, gives:

$$\boxed{\mathcal{P}_c - \mathcal{P}_p = S(0) - K \cdot \mathcal{D}_{T,0}^{-1}(g_y)}$$

Parity uses only the payoff identity and the share's zero expected NPV, so it holds whatever model describes the share price under $\mathbb{Q}$.

Example 2: CHEME-5660-L9a-Example-BSM-Premium-Fall-2026.ipynb

Optional Advanced Material

Contingent claims: advanced/contingent_claims/
CHEME-5660-L9a-Advanced-ContingentClaims-BSM-Fall-2026.ipynb

Where do the BSM premiums come from? Replication on a one-step lattice gives the risk-neutral probability and the state prices that price event contracts. The call is an asset-or-nothing claim minus K digital claims. Marginal utility then shows where the bet's risk premium goes.

Volatility skew: advanced/spxw_volatility_skew/
CHEME-5660-L9a-Advanced-SPXW-Volatility-Skew-Fall-2026.ipynb

Does one volatility price every strike? We invert a synthetic SPXW-style chain of European, cash-settled index options for implied volatility.

Notebooks: lectures/week-9/L9a/advanced, also linked from the lecture notebook.

Summary

Today we introduced derivatives, read the payoff and profit of call and put contracts, and priced European contracts.

Key takeaways

- Derivatives derive their value from underlying assets: common types are futures, forwards, swaps, and options.
- Option payoffs are set by the strike and the position: the premium turns payoff into profit and sets the breakeven.
- The premium makes the expected NPV zero under $\mathbb{Q}$: the BSM formulas compute it for European contracts, and parity links calls and puts.